LEVEL-III

1. $\int_{-\pi}^{\pi}(\cos ax-\sin bx)^2\,dx$ **(where a, b are integers)=**
1) $-\pi$ 2) 0 3) π 4) 2π

2. $\int_{0}^{\pi/2}\left(\frac{x}{\sin x}\right)^2 dx=$
1) $\pi\log 2$ 2) $-\frac{\pi}{2}\log 2$ 3) $\frac{\pi}{2}\log 2$ 4) $-\pi\log 2$

3. **If** $f(y)=e^y, g(y)=y; y>0$ **and** $F(t)=\int_0^t f(t-y)g(y)\,dy$, **then**
1) $F(t)=e^t-(1+t)$ 2) $F(t)=te^t$
3) $F(t)=te^{-t}$ 4) $F(t)=1-e^{-t}(1+t)$

4. **If** $2f(x)+3f\left(\frac{1}{x}\right)=\frac{1}{x}-2,\ x\neq 0$, **then** $\int_1^2 f(x)$ **dx is equal to**
1) $\frac{-2}{5}\log 2+\frac{1}{2}$ 2) $\frac{-2}{5}\log 2-\frac{1}{2}$
3) $\frac{2}{5}\log 2+\frac{1}{2}$ 4) $-\frac{2}{5}\log 2$

5. $\int_{-1}^{1}[x\sin\pi x]\,dx=$
1) 0 2) 1 3) 2 4) 3

6. $\int_{-3\pi/2}^{-\pi/2}\left((x+\pi)^3+\cos^2(x+3\pi)\right)dx=$
1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}-1$ 3) $\frac{\pi^4}{32}$ 4) $\frac{\pi^4}{32}+\frac{\pi}{2}$

7. $\int_{-1}^{3/2}|x\sin\pi x|\,dx=$
1) $\frac{3}{\pi}-\frac{1}{\pi^2}$ 2) $\frac{3}{\pi}+\frac{1}{\pi^2}$ 3) $\frac{2}{\pi}+\frac{1}{\pi^2}$ 4) $\frac{3}{\pi}+\frac{1}{\pi}$

8. $\int_{-1/2}^{1/2}\sqrt{\left(\frac{x+1}{x-1}\right)^2+\left(\frac{x-1}{x+1}\right)^2-2}\ dx=$
1) $8\log\frac{4}{3}$ 2) $8\log\frac{3}{4}$ 3) $4\log\frac{4}{3}$ 4) 0

9. **The value of** x **in the interval** $(-\pi,0)$ **satisfying** $\sin x+\int_x^{2x}\cos 2t\ dt=0$ **is**
1) $-\frac{\pi}{2}$ 2) $-\frac{\pi}{3}$ 3) $-\frac{\pi}{4}$ 4) $-\frac{\pi}{6}$

10. $\int_4^{10}([x-4]+[10-x])dx=$ **(where** $[x]$ **is the largest integer not exceeding x)**
1) 26 2) 10 3) 30 4) 20

11. **The greatest value of** $f(x)=\int_1^x|t|\,dt$ **on the interval** $\left[-\frac{1}{2},\frac{1}{2}\right]$ **is :**
1) $\frac{3}{8}$ 2) $-\frac{3}{8}$ 3) $-\frac{1}{2}$ 4) $\frac{1}{2}$

12. **Let** $f(x)=\int_0^x\frac{\cos t}{t}dt\quad(x>0)$ **then for** $x=(2n+1)\frac{\pi}{2}; f(x)$ **has :**
1) maxima when $n=0,2,4,6,\ldots\ldots\ldots\ldots$
2) minima when $n=0,2,4,6,\ldots\ldots\ldots\ldots$
3) nethir maxima nor minima when $n=-1,-3,-5,\ldots\ldots\ldots\ldots$
4) in sufficient data

13. $\int_{\pi}^{5\pi/4}\frac{\text{Sin}2x.dx}{\text{Cos}^4x+\text{Sin}^4x}=$
1) $\frac{5\pi}{4}$ 2) $\frac{\pi}{2}$ 3) π 4) $\pi/4$

14. $\int_0^{\pi/4}\frac{x^2dx}{(x\text{Sin}x+\text{Cos}x)^2}=$
1) $\frac{4+\pi}{4-\pi}$ 2) $\frac{4+\pi}{2(4-\pi)}$
3) $\frac{4-\pi}{4+\pi}$ 4) $2\left[\frac{4-\pi}{4+\pi}\right]$

15. [y] is the integral part of y, then $\int_{\pi/2}^{3\pi/2}[2\sin x]dx =$

1) $-\pi$ 2) 0 3) $-\pi/2$ 4) $\pi/2$

16. If $(1+x)^{2n}=\sum_{r=0}^{2n}C_r x^r$, then $\int_0^2 (C_0+C_2+C_4+........C_{2n})$ dx equals

1) 2^{2n-1} 2) 2^{3n} 3) 4^n 4) 4^{n-1}

17. The value of $\int_0^{\pi/2}\frac{\cos 3x+1}{2\cos x-1}$ dx is

1) 2 2) 1 3) 1/2 4) 0

18. $\int_0^{7\frac{1}{2}}[x-1]dx =$ Where [x] denotes the greatest integer less than or equal to x

1) 5/2 2) 1 3) 5/4 4) 17

19. A line tangent to the graph of the function y = $f(x)$ at the point $x=a$ forms an angle $\pi/3$ with the axis of abscissas and angle $\pi/4$ at the point $x=b$ then $\int_a^b f''(x)dx$ is (f'' is assumed to be a continuous function)

1) 0 2) $1-\sqrt{3}$ 3) $\sqrt{2}-1$ 4) $\sqrt{3}-1$

20. For y = f(x) = $\int_0^x 2|t|$ dt, the tangent lines parallel to the bisector of the first quadrant angle are

1) $y=x\pm\frac{1}{4}$ 2) $y=x\pm\frac{3}{2}$

3) $y=x\pm\frac{1}{2}$ 4) y =x +2

21 Equation of tangent to $y=\int_{x^2}^{x^3}\frac{dt}{\sqrt{1+t^2}}$ at $x=1$ is

1) $y=x\sqrt{2}-1$ 2) $y\sqrt{2}=x-1$

3) $y=x\sqrt{2}+1$ 4) $y\sqrt{2}=x+1$

22. Let $f(x)$ be a function satisfying $f'(x)=f(x)$ with $f(0)=1$ and $g(x)$ be a function that satisfies $f(x)+g(x)=x^2$. Then, the value of the integral $\int_0^1 f(x)g(x)\,dx$ is

1) $e+\frac{e^2}{2}-\frac{3}{2}$ 2) $e-\frac{e^2}{2}-\frac{3}{2}$

3) $e+\frac{e^2}{2}+\frac{5}{2}$ 4) $e-\frac{e^2}{2}-\frac{5}{2}$

23. Let (a,b) and (λ,μ) be two points on the curve $y=f(x)$. If the slope of the tangent to the curve at (x,y) be $\phi(x)$, then $\int_a^\lambda \phi(x)dx$ is

1) $\lambda-a$ 2) $\mu-b$

3) $\lambda+\mu-a-b$ 4) $\lambda+\mu-b$

24. If $f(x)=\int_0^x\frac{dt}{\{f(t)\}^2}$ and $\int_0^2\frac{dt}{\{f(t)\}^2}=\sqrt[3]{6}$ then $f(9)=$

1) 0 2) 1

3) 2 4) 3

25. If $A=\int_0^\pi\frac{\cos x}{(x+2)^2}dx$ then $\int_0^{\pi/2}\frac{\sin 2x}{x+1}dx$ is equal to

1) $A-\frac{1}{2}-\frac{1}{\pi+2}$ 2) $\frac{1}{2}+\frac{1}{\pi+2}-A$

3) $\frac{1}{\pi+2}-A$ 4) $1+\frac{1}{\pi+2}-A$

26. If $I_1=\int_0^{\pi/2}f(\sin 2x)\sin x\,dx$, and $I_2=\int_0^{\pi/4}f(\cos 2x)\cos x\,dx$ then $\frac{I_1}{I_2}=$

1) $\sqrt{2}$ 2) $2\sqrt{2}$

3) $3\sqrt{2}$ 4) $4\sqrt{2}$

27. The value of $\int_0^4 \{\sqrt{x}\}dx$, where $\{.\}$ denotes the fractional part function, is

1) $\frac{1}{3}$ 2) 1 3) $\frac{5}{3}$ 4) $\frac{7}{3}$

28. $\int_{-\pi/4}^{n\pi-\pi/4} |\sin x+\cos x| dx \ (n \in N)$ is

1) 0 2) $2n$ 3) $2\sqrt{2}n$ 4) $\sqrt{2}$

29. $\int_{-\pi/4}^{\pi/4} \frac{e^x.\sec^2 x}{e^{2x}-1} dx$ is equal to

1) 0 2) 2 3) e 4) 1

30. The value of the integral $I=\int_1^{\infty} \frac{x^2-2}{x^3\sqrt{x^2-1}} dx$ is

1) 0 2) $\frac{2}{3}$ 3) $\frac{4}{3}$ 4) 1

31. If $a_n=\int_0^{\pi/2} \frac{\sin^2 nx}{\sin x} dx$ then a_2-a_1, a_3-a_2, a_4-a_3 are in

1) AP 2) GP 3) HP 4) AGP

32 If $\int_{1/2}^{2} \frac{1}{x} \operatorname{cosec}^{101}\left(x-\frac{1}{x}\right) dx = k$ then the value of k is

1) 1 2) $\frac{1}{2}$ 3) 0 4) $\frac{1}{101}$

33. The value of the definite integral $\int_0^1 \frac{x\,dx}{x^3+16}$ lies in the interval $[a,b]$. Then smallest such interval is

1) $\left[0,\frac{1}{17}\right]$ 2) $[0,1]$ 3) $\left[0,\frac{1}{27}\right]$ 4) $\left[\frac{1}{17},1\right]$

34. Let $\int_0^x \left(\frac{bt\cos 4t-a\sin 4t}{t^2}\right)dt=\frac{a\sin 4x}{x}$ then a and b are given by

1) $a=\frac{1}{4}, b=1$ 2) $a=2, b=2$

3) $a=-1, b=4$ 4) $a=2, b=4$

35. The integral $\int_0^{\pi} \sqrt{1+4\sin^2 x/2-4\sin x/2}\, dx =$

[JEE MAINS -2014]

1) $\pi-4$ 2) $\frac{2\pi}{3}-4-4\sqrt{3}$

3) $4\sqrt{3}-4$ 4) $4\sqrt{3}-4-\pi/3$

36. If $I=\int_{-\pi/6}^{\pi/6} \frac{\pi+4x^5}{1-\sin\left(|x|+\frac{\pi}{6}\right)} dx$, then I equals to

1) 4π 2) $2\pi+\frac{1}{\sqrt{3}}$

3) $2\pi-\sqrt{3}$ 4) $4\pi+\sqrt{3}-\frac{1}{\sqrt{3}}$

37 If $I=\int_0^{\pi/2} \cos^n x\sin^n x\,dx=\lambda\int_0^{\pi/2} \sin^n x\,dx$ then $\lambda=$

1) 2^{-n+1} 2) 2^{-n-1} 3) 2^{-n} 4) 2^{-1}

38. If for $K \in N$

$\frac{\sin 2kx}{\sin x}=2[\cos x+\cos 3x+.....+\cos(2k-1)x]$

Then the value of $I=\int_0^{\pi/2} \sin 2kx.\cot x\,dx$ is

1) $\frac{-\pi}{2}$ 2) 0 3) $\frac{\pi}{2}$ 4) π

39. Find the interval in which

$F(x)=\int_{-1}^{x} (e^t-1)(2-t)dt$, $(x>-1)$ is increasing

1) $[3,5]$ 2) $[1,3]$ 3) $[0,3]$ 4) $[0,2]$

40. If $f(\pi)=2$ and

$\int_0^{\pi} (f(x)+f''(x))\sin x\,dx=5$ then $f(0)$ is equal to

1) 7 2) 3 3) 5 4) 1

41. If $f:R\to R$ is a differentiable function and $f(1)=4$. Then the value of $\lim_{x\to 1} \frac{\int_4^{f(x)} 2t\,dt}{x-1}$ is

1) $8f'(1)$ 2) $4f'(1)$ 3) $2f'(1)$ 4) $f'(1)$

42. If $f(x)=\int_{x^2}^{x^2+1} e^{-t^2}dt,$ **then** $f(x)$ **increases in**

1) $(-2,2)$ 2) no value of x

3) $(0,\infty)$ 4) $(-\infty,0)$

43. $\int_0^{\frac{\pi}{4}}\left[\sin x+\left[\cos x+\left[\tan x+\left[\sec x\right]\right]\right]\right]dx=$

1) $\frac{\pi}{3}$ 2) $\frac{\pi}{4}$ 3) $\frac{2\pi}{3}$ 4) $\frac{3\pi}{4}$

44. $\int_{-2}^{3}\left[|x|\right]d|x|=$

1) 0 2) 1 3) 2 4) 4

45. If = $y=\{x\}^{[x]}$ $then\int_0^3 y\,dx$

1) $\frac{1}{6}$ 2) $\frac{5}{6}$ 3) $\frac{7}{6}$ 4) $\frac{11}{6}$

46. If $f(2-x)=f(2+x), f(4+x)=f(4-x)$ **and** $\int_0^2 f(x)dx=5$ $then$ $\int_0^{50} f(x)dx$

1) 25 2) 50 3) 75 4) 125

47. The period of f(x) is 1 and $f\left(\frac{1}{2}\right)=\frac{1}{2}$**. If** $\phi(x)=\int_0^x f(t+n)dt$ $n\in N$ **then** $\phi^1(3/2)=$

1) $\frac{1}{2}$ 2) 1 3) 3/2 4) 2

48. If x = $\int_0^y\frac{dt}{\sqrt{1+9t^2}}$ **and** $\frac{d^2y}{dx^2}=ay$ **then a =**

1) 7 2) 8 3) 9 4) 6

49. If $f(x)=\int_{x^2}^{x^2+1}e^{-t^2}dt$ **find interval in which** $f(x)$ **in increasing**

1) $(0\ \infty)$ 2) $(-\infty\ 0)$ 3) $(-2\ 2)$ 4) (0 2)

50. If $\int_0^\infty\frac{\sin x}{x}dx=\frac{\pi}{2}$ **then** $\int_0^\infty\frac{\sin^3 x}{x}dx$ **is equal to**

1) $\frac{\pi}{2}$ 2) $\frac{\pi}{4}$ 3) $\frac{\pi}{6}$ 4) $3\frac{\pi}{2}$

51. The true set of values of 'a' for which the inequality $\int_a^0\left(3^{-2x}-2.3^{-x}\right)dx\ \geq 0$ **is true**

1) $[0\ 1]$ 2) $(-\infty,-1]$ 3) $[0,\infty)$

4) $(-\infty,1]\cup[0,\infty)$

52. The value of the definite integral $\int_0^{\pi/2}\sin|2x-\alpha|\,dx$ **where** $\alpha\in[0,\pi]$

1) 1 2) $\cos\alpha$ 3) $\frac{1+\cos\alpha}{2}$ 4) $\frac{1-\cos\alpha}{2}$

53. $\int_0^{\pi/2}\frac{\sin x}{x}$ **lies between**

1) 0 and 1 2) -1 and 1 3) $1 and \frac{\pi}{2}$ 4) -1 and 0

54. Let f be a continuous function satisfying

$$f^1(\log x)=\begin{cases}1 & for\ 0<x\leq 1\\ x & for\ x>1\end{cases}\ and\ f(0)=0$$

Then f(x) can be defined as

1) $f(x)=\begin{cases}1 & if\ x\leq 0\\ 1-e^x & if\ x>0\end{cases}$

2) $f(x)=\begin{cases}1 & if\ x\leq 0\\ e^x-1 & if\ x>0\end{cases}$

3) $f(x)=\begin{cases}x & if\ x<0\\ e^x & if\ x>0\end{cases}$

4) $f(x)=\begin{cases}x & if\ x\leq 0\\ e^x-1 & if\ x>0\end{cases}$

55 If $g(x)=\int_0^x\cos 4t\,dt,$ then $g(x+y)=$ **(AIEEE-2012)**

1) $\frac{g(x)}{g(\pi)}$ 2) $g(x)+g(\pi)$

3) $g(x)-g\left(\frac{\pi}{3}\right)$ 4) $g(x).g(\pi)$

56. **Statment-I:** The value of the integral $\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$ is equal to $\pi/6$

Statement-II: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

(JEE-MAINS-2013)

1) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I.
2) Statement-I is True, Statement-II is True; Statement-II is NOT a correct explanation for Statement-I.
3) Statement-I is True, Statement-II is False.
4) Statement-I is False, Statement-II is True

57. If $I_1 = \int_{1/e}^{\tan x} \frac{t}{1+t^2}dt$ and $I_2 = \int_{1/e}^{\cot x} \frac{dt}{t(1+t^2)}$ then the value of $I_1 + I_2$ is

1) 1/2 2) 1
3) e/2 4) (1/2)(e+1/e)

58. $\int_0^{\pi} \frac{x\sin 2x \sin\left(\frac{\pi\cos x}{2}\right)}{2x-\pi}dx =$

1) $\frac{2}{\pi^2}$ 2) $\frac{8}{\pi^2}$
3) $\frac{3}{\pi^2}$ 4) $\frac{4}{\pi^2}$

59. If $f(x) = Ax^2 + Bx$ satisfies the conditions $f'(1) = 8$ and $\int_0^1 f(x)dx = \frac{8}{3}$, then

1) $A = 1,\ B = -4$ 2) $A = 2,\ B = 4$
3) $A = -2,\ B = 4$ 4) $A = -2,\ B = -4$

60. If $f(x) = ae^{2x} + be^x + cx$,satisfies the conditions $f(0) = -1$, $f'(\log 2) = 31$, , $\int_0^{\log 4} (f(x) - cx)dx = \frac{39}{2}$, then

1) a=5,b=6,c=3 2) a=5,b=-6,c=3
3) a=-5,b=6,c=3 4) a= 1, b = 2, c=3

LEVEL - III - KEY

1) 4	2) 1	3) 1	4) 1	5) 1	6) 1
7) 2	8) 3	9) 2	10) 3	11) 2	12) 1
13) 4	14)3	15)3	16) 3	17) 2	18) 4
19)2	20)1	21)2	22) 2	23) 2	24) 4
25) 2	26) 1	27) 4	28) 3	29) 1	30) 1
31) 3	32) 3	33) 1	34) 1	35) 4	36) 1
37) 3	38) 3	39) 4	40) 2	41) 1	42) 4
43) 2	44) 3	45) 4	46) 4	47) 3	48) 3
49) 2	50) 2	51) 4	52) 1	53)3	54)4
55)2	56)4	57)2	58)2	59)2	60) 2

LEVEL - III - HINTS

1. $I = \int_{-\pi}^{\pi}\left(\cos^2 ax + \sin^2 bx\right)dx - 2\int_{-\pi}^{\pi}\cos ax \sin bx\,dx$

$= 2\int_0^{\pi}\left(\cos^2 ax + \sin^2 bx\right)dx - 0$

$= \int_0^{\pi}\left(1+\cos 2ax + 1 - \cos 2bx\right)dx \ = 2\pi + 0 = 2\pi$

2. Given $= \int_0^{\frac{\pi}{2}} x^2 \cos ce^2 x dx$ and using byparts

3. $F(t) = \int_0^t f(t-y)g(y)dy$

$= \int_0^t e^{t-y}.y\,dy$ (integration by parts) $= e^t - t - 1$

4. we have, $2f(x) + 3f\left(\frac{1}{x}\right) = \frac{1}{x} - 2$

$\Rightarrow 2f\left(\frac{1}{x}\right) + 3f(x) = x - 2$

solving the above two equations we get

$f(x) = \frac{-2}{5x} + \frac{3x}{5} - \frac{2}{5}$

5. $\int_{-1}^{1}[x\sin\pi x]dx = 2\int_0^1 [x\sin\pi x]dx$

($\because$ The integral is an even function of x)

now $\Rightarrow 0 \le \sin\pi x \le 1 \Rightarrow [x\sin\pi x] = 0$.

6. sub $x + \pi = t$

7. Given $= \int_{-1}^{1}|xSin\pi x|dx + \int_1^{\frac{3}{2}}|xSin\pi x|dx$

8. $\int_{-1/2}^{1/2}\sqrt{\left(\frac{x+1}{x-1} - \frac{x-1}{x+1}\right)^2}\,dx \ = \int_{-1/2}^{1/2}\left|\frac{x+1}{x-1} - \frac{x-1}{x+1}\right|dx$

9. $\sin x+\left(\frac{\sin 2x}{2}\right)_x^{2x}=0$;

$\sin x+\frac{\sin 4x-\sin 2x}{2}=0$

$\sin x+\cos 3x\sin x=0$

10. $\int_4^{10}\{[x-4]+[10-x]\}dx=\int_4^{10}\{[x]-4+10+[-x]\}dx$

$=\int_4^{10}\{[x]+[-x]\}dx+\int_4^{10}6dx = \int_4^{10}-1dx+\int_4^{10}6dx$

11 $f'(x)=|x|\geq 0$; $f(x)$ is increasing in $\left[-\frac{1}{2},\frac{1}{2}\right]$

max $=f\left(\frac{1}{2}\right)=\int_1^{1/2}|t|dt$

12 $f'(x)=\frac{\cos x}{x}=0$;

$f''(x)=\frac{-x\sin x-\cos x}{x^2}<0$

13. Dividing by $\cos^4 x$ and $Tan^2x=t$

14. Given $=\frac{x\cos x}{(x\sin x+\cos x)^2}\times\frac{x}{\cos x}$ and using by parts

15. $=\int_{\frac{\pi}{2}}^{\frac{5\pi}{6}}[2S\text{in}\,x]dx+\int_{\frac{5\pi}{6}}^{\pi}[2S\text{in}\,x]dx+\int_{\pi}^{\frac{7\pi}{6}}[2S\text{in}\,x]dx+\int_{\frac{7\pi}{6}}^{\frac{3\pi}{2}}[2S\text{in}\,x]dx$

16. $(1+x)^{2n}={}^{2n}C_0+{}^{2n}C_1x^1+{}^{2n}C_2x^2+........+{}^{2n}C_{2n}x^{2n}$

Putting x =1, -1 we get

$2^{2n}=C_0+C_1+C_2+........+C_{2n}$

where $\left(C_0={}^{2n}C_0\right)$ and

$0=C_0-C_1+C_2+.........+C_{2n}$ on adding

$2^{2n-1}=C_0+C_2+C_4+.........+C_{2n}$

$\therefore\quad \int_0^2 2^{2n-1}dx=\int_0^2(C_0+C_2+C_4+..........+C_{2n})dx$

$\Rightarrow\quad \int_0^2(C_0+C_2+C_4+=.........+C_{2n})dx=(2^{2n-1})(2)=4^n$

17. $\int_0^{\pi/2}\frac{\cos 3x+1}{2\cos x-1}dx==\int_0^{\pi/2}\frac{\cos 3x-\cos 3\pi/3}{2(\cos x-\cos\pi/3)}dx$

$=\int_0^{\pi/2}(1+\cos 2x+\frac{1}{2}+\cos x)\,dx-,$

$\sin ce\cos 3x=4\cos^3 x-3\cos x$

$=\frac{3\pi}{4}+1-\frac{3\pi}{4}=1$

18. If f(x) = [x-1] then given =

$\int_0^1 f(x)dx+\int_1^2 f(x)dx+.......+\int_6^7 f(x)dx+\int_7^{7\frac{1}{2}} f(x)dx$

19. $\int_a^b f''(x)dx=f'(b)-f'(a)$ Given

$f'(b)=\tan\frac{\pi}{4}=1$ and $f'(a)=\tan\frac{\pi}{3}=\sqrt{3}$

20. $f(x)=\int_0^x 2|t|dt\Rightarrow f^1(x)=2|x|$

Now $f^1(x)=1\Rightarrow 2|x|=1\Rightarrow x=\pm\frac{1}{2}$

$\therefore$ The equation of the tangents at $\left(\frac{1}{2},\frac{1}{4}\right)$ and $\left(\frac{-1}{2},\frac{-1}{4}\right)$ are

$y=x-\frac{1}{4}$ and $y=x+\frac{1}{4}$

21. When $x=1, y=0,$

Slope$=\frac{1}{\sqrt{1+x^6}}(3x^2)-\frac{1}{\sqrt{1+x^4}}(2x)=\frac{1}{\sqrt{2}}$

$\therefore\sqrt{2}y=x-1$

22. $f^1(x)=f(x)\Rightarrow f(x)=e^x$

$f(x)+g(x)=x^2\Rightarrow g(x)=x^2-e^x$

$\int_0^1 f(x)g(x)dx=\int_0^1(x^2e^x-e^{2x})dx$

23. $f'(x)=\phi(x)$

$\int_a^{\lambda}\phi(x)dx=\int_a^{\lambda}f'(x)dx=(f(x))_a^{\lambda}=f(\lambda)-f(a)$

But $b=f(a)$ and $\mu=f(\lambda)$

24. $\because f(x)=\int_0^x \frac{dt}{[f(t)]^2} \Rightarrow f^1(x)=\frac{1}{[f(x)]^2}$

$\Rightarrow f^1(x)[f(x)]^2=1 \Rightarrow \frac{[f(x)]^3}{3}=x+c$

Put $x=2 \Rightarrow \frac{[f(2)]^3}{3}=2+c \Rightarrow \frac{6}{3}=2+c$

$\left[\therefore f(2)=\sqrt[3]{6}\right] \quad \Rightarrow c=0$

$\Rightarrow f(x)=(3x)^{\frac{1}{3}} \therefore f(9)=3$

25. $\because A=\int_0^{\pi}\frac{\cos x}{(x+2)^2}dx$

$\Rightarrow A=\left\{\cos x\left(\frac{-1}{x+2}\right)\right\}_0^{\pi}-\int_0^{\pi}(-\sin x)\left(\frac{-1}{x+2}\right)dx$

$\Rightarrow A=\frac{1}{\pi+2}+\frac{1}{2}-\int_0^{\pi}\frac{\sin x}{x+2}dx \quad [\text{Put } x=2t]$

$A=\frac{1}{\pi+2}+\frac{1}{2}-\int_0^{\pi/2}\frac{\sin 2x}{x+1}dx$

$\int_0^{\pi/2}\frac{\sin 2x}{x+1}dx=\frac{1}{\pi+2}+\frac{1}{2}-A$

26. $I_1+I_1=\int_0^{\pi/2} f(\sin 2x)(\sin x+\cos x)\,dx$

$=\sqrt{2}\int_0^{\pi/4} f(\sin 2x)\cos\left(x-\frac{\pi}{4}\right)dx$

$=\sqrt{2}\int_{-\pi/4}^{\pi/4} f(\cos 2t)\cos t \text{ dt where } t=x-\frac{\pi}{4}$

31. $\int_0^4\{\sqrt{x}\}dx=\int_0^4\left(\sqrt{x}-[\sqrt{x}]\right)dx$

$=\int_0^4\sqrt{x}\,dx-\int_0^4[\sqrt{x}]dx=\frac{16}{13}-\left[\int_0^{1^2}[x]dx+\int_{1^2}^{2^2}[\sqrt{x}]dx\right]$

$=\frac{16}{3}-\left[0+1.(2^2-1^2)\right]=\frac{16}{3}-3=\frac{7}{3}$

27. $\int_0^4\{\sqrt{x}\}dx=\int_0^4\left(\sqrt{x}-[\sqrt{x}]\right)dx$

$=\int_0^4\sqrt{x}\,dx-\int_0^4[\sqrt{x}]dx=\frac{16}{13}-\left[\int_0^{1^2}[x]dx+\int_{1^2}^{2^2}[\sqrt{x}]dx\right]$

$=\frac{16}{3}-\left[0+1.(2^2-1^2)\right]=\frac{16}{3}-3=\frac{7}{3}$

28. $I=\int_{-\pi/4}^{n\pi-\pi/4}|\sin x+\cos x|dx=\sqrt{2}\int_{-\pi/4}^{n\pi-\pi/4}\left|\sin\left(x+\frac{\pi}{4}\right)\right|dx$

Put $x+\frac{\pi}{4}=t \quad dx=dt$;

$I=\sqrt{2}\int_0^{n\pi}|\sin t|dt=n\sqrt{2}\int_0^{\pi}|\sin t|dt$

29. Let $I=\int_{-\pi/4}^{\pi/4}\frac{e^x.\sec^2 x}{e^{2x}-1}dx$

If $f(x)=\frac{e^x\sec^2 x}{e^{2x}-1}$ then $f(-x)=-f(x)$

$\therefore I=0$

30. $I=\int_1^{\infty}\frac{x^2-2}{x^3\sqrt{x^2-1}}dx \qquad$ Put $x=\frac{1}{t} \quad dx=-\frac{1}{t^2}dt$

$I=\int_0^1\frac{1-2t^2}{\sqrt{1-t^2}}dt$ Put $t=\sin\theta$;

$I=\int_0^{\pi/2}\cos 2\theta d\theta=0$

31. $a_n-a_{n-1}=\int_0^{\pi/2}\frac{\sin^2 nx-\sin^2(n-1)x}{\sin x}dx$

$=\int_0^{\pi/2}\frac{\sin(2n-1)x\,\sin x}{\sin x}dx$

$=\left[-\frac{1}{2n-1}\cos(2n-1)x\right]_0^{\pi/2}=\frac{1}{2n-1}$

$\therefore a_2-a_1=\frac{1}{3}, a_3-a_2=\frac{1}{5}, a_4-a_3=\frac{1}{7}$

32. $\because \int_{1/2}^{2} \frac{1}{x} \cos ec^{101}\left(x-\frac{1}{x}\right) dx = k$

Put $x=\frac{1}{t}$ $\quad dx=-\frac{1}{t^2}dt$

$\Rightarrow \int_{2}^{1/2} t \cos ec^{101}\left(\frac{1}{t}-t\right) \times \frac{-1}{t^2} dt = k$

$= \int_{1/2}^{2} \frac{1}{t} \cos ec^{101}\left(\frac{1}{t}-t\right) dt = k \quad \Rightarrow -k=k$

$2k=0 \Rightarrow k=0$

33. We have

$(1-0)\left(\frac{0}{0^3+16}\right) \le \int_0^1 \frac{x}{x^3+16} dx \le (1-0)\left(\frac{1}{1^3+16}\right)$

(by property) $\quad 0 \le \int_0^1 \frac{x\,dx}{x^3+16} \le \frac{1}{17}$

34. $\because \int_0^x \left(\frac{bt\cos 4t - a\sin 4t}{t^2}\right) dt = \frac{a\sin 4x}{x}$

Diff.both sides wrt 'x'

$\frac{bx\cos 4x - a\sin 4x}{x^2} = \frac{a(4x\cos 4x - \sin 4x)}{x^2}$

on comparing $b=4a$ $\quad \because a=\frac{1}{4}, b=1$

35. $I = \int_0^\pi \sqrt{1+4\sin^2 x/2 - 4\sin x/2}$

$I = \int_0^\pi |1-2\sin x/2|\,dx$; $4\sqrt{3}-4-\pi/3$

36. As $\dfrac{4x^5}{1-\sin\left(|x|+\frac{\pi}{6}\right)}$ is an odd function, and

$\dfrac{\pi}{1-\sin\left(|x|+\frac{\pi}{6}\right)}$ is an even function, we get

$I = 2\pi \int_0^{\pi/6} \frac{dx}{1-\sin\left(x+\frac{\pi}{6}\right)}$ Put $x+\frac{\pi}{6}=t$

$= 2\pi \int_{\pi/6}^{\pi/3} \frac{dt}{1-\sin t} = 4\pi$

37. $I = \frac{1}{2^n} \int_0^{\pi/2} (2\sin x\cos x)^n dx$

$I = \frac{1}{2^n} \int_0^{\pi/2} (\sin 2x)^n dx$ Put $2x=t$

$= \frac{1}{2^n} \int_0^\pi \sin^n \theta . \frac{1}{2} d\theta$

$I = \frac{1}{2^{n+1}} \int_0^{\pi/2} \left([\sin\theta]^n + \sin(\pi-\theta)^n\right) d\theta$

$I = \frac{1}{2^n} \int_0^{\pi/2} \sin^n \theta$ Prove by

$\left[\int_0^{2a} f(x)\,dx = \int_0^a [f(x)+f(2a-x)]dx\right.$

38. $I = \int_0^{\pi/2} \frac{\sin 2kx}{\sin x} \cos x\,dx$ From given identity

$I = \int_0^{\pi/2} 2[\cos x + \cos 3x + ...+\cos(2k-1)x].\cos x\,dx$

$= \int_0^{\pi/2} [1+\cos 2x + \cos 4x + \cos 2x +$

$.....\cos 2kx + \cos(2k-2x)]dx$

$= \left[x+\sin 2x + \frac{1}{2}\sin 4x + ...+\frac{1}{2k}\sin 2kx\right]_0^{\pi/2} = \frac{\pi}{2}$

39. Use Newton Leibnitz rule,

$F'(x) = (e^x-1)(2-x) > 0 \Rightarrow x \in [0,2]$

40. $\int_0^\pi f(x)\sin x\,dx + \int_0^\pi f''(x)\sin x\,dx =$

I II II I

Use byparts

$\Rightarrow f(\pi)+f(0)=5 \qquad \therefore f(0)=3$

41. $\lim_{x\to 1} \frac{\int_4^{f(x)} 2t\,dt}{x-1}$ (Using L' Hospital' rule)

$\Rightarrow 2f(1).f'(1) \;=8f'(1)$

42.. $f'(x)=2x\left[\frac{1}{e^{(x^2+1)^2}}-\frac{1}{e^{x^4}}\right]>0$ if $x<0$

43. $I=\int_0^{\pi/4}\left[\sin x+\left[\cos x+\left[\tan x+1\right]\right]\right]$

$=\int_0^{\pi/4}\left[\sin x+\left[\cos x+1\right]\right]=\int_0^{\pi/4}1.dx=\frac{\pi}{4}$

44. $I=-\int_{-2}^{0}[-x]dx+\int_0^3(x)\,dx$

$=-\int_0^2[x]dx+\int_0^2[x]\,dx+\int_2^3[x]\,dx$

45. $I=\int_0^3(x-[x])^{[x]}dx=\int_0^1 x^0\,dx+\int_1^2(x-1)^1\,dx+\int_2^3(x-2)^2\,dx$

46. Replace x by x-2 we get f(4-x) = f(x) hence f(4+x) = f(x) $\Rightarrow$ period f(x) is 4

$I=\int_0^{48}f(x)\,\text{dx}+\int_{48}^{50}f(x)\,dx$

$=12\int_0^2 f(x)\,\text{dx}+\int_0^2 f(x+48)\,dx\quad=12[5+5]$

47. Period of f(x) = 1 $\Rightarrow f(x)=f(1+x)$

for $n\in N\;\; f(x+n)=f(x)$

$\therefore\phi^1(x)=f(x+n)=f(x)$

$\Rightarrow\phi^1(3/2)=f\left(\frac{3}{2}\right)=f\left(1+\frac{1}{2}\right)=f\left(\frac{1}{2}\right)$

48. $\frac{dx}{dy}=\frac{1}{\sqrt{1+9y^2}}\Rightarrow\frac{d^2y}{dx^2}=9y$

49. $f^1(x)>0\Rightarrow e^{-(x^2+1)^2}(2x)-e^{-(x^2)^2}(2x)>0$

$\Rightarrow 2xe^{-(x^2+1)^2}\left[1-e^{2x^2+1}\right]>0$

$for\; x<0, 1-e^{2x^2+1}<0$

50. $I=\int_0^\infty\frac{3\sin x-\sin 3x}{4x}\,dx\;=\frac{3}{4}\frac{\pi}{2}-\frac{1}{4}\int_0^\infty\frac{\sin 3x}{x}\,dx$

$=\frac{3\pi}{8}-\frac{1}{4}\int_0^\infty\frac{\sin t}{3.\frac{t}{3}}\,dt\;=\frac{3\pi}{8}-\frac{1}{4}\;\frac{\pi}{2}$

51. $\int_a^0 3^{-x}\left(3^{-x}-2\right)dx\ge 0, put 3^{-x}=t$

$\left(\frac{t^2}{2}-2t\right)_1^{3^{-a}}\ge 0\;\Rightarrow 3^{-2a}-4\;3^{-a}+3\ge 0$

52. $\int_0^{\pi/2}\sin|2x-\alpha|\,dx$

$=\frac{1}{2}\int_{-\propto}^{\pi/\propto}\sin|t|\,dt$

$=\frac{1}{2}\int_{-\alpha}^{0}(-\sin t)\,dt+\frac{1}{2}\int_0^{\pi-\propto}\sin t\,dt$

$=\frac{1}{2}[\cos t]_{-\alpha}^{0}-\frac{1}{2}[\cot t]_0^{\pi-\alpha}=1$

53. $\sin x<x<\tan x$ for $x\in\left(0\,\frac{\pi}{2}\right)\Rightarrow 1>\frac{\sin x}{x}>\cos x$

54. Put $\log x=t\Rightarrow\frac{1}{x}\,dx=dt$

$\Rightarrow dx=e^t dt$ Again

For $x>1\;\; f^1(t)=e^t, t>o$ for $0<x\le 1$

Integrating $f(t)=e^t+c$ $\quad f^1(\log x)=1$

$f(0)=e^0+c\Rightarrow c=-1$ $\quad f^1(t)=1, t\le 0$

$f(t)=e^t-1\;\; for\;\; t>o$ $\quad f^1(t)=t+c$

$\therefore f(x)=e^x-1\;\; for\;\; x>o$

55. $\Rightarrow g(x+\pi)=\int_{t=0}^{t=x+y}\cos 4t\;dt$

56. $I=\int_{\pi/3}^{\pi/3}\frac{dx}{1+\sqrt{\tan x}}$ ......(1)

$\therefore I=\int_{\pi/3}^{\pi/3}\frac{\sqrt{\tan x}dx}{1+\sqrt{\tan x}}$ ......(2) $\Rightarrow 2I=\int_{\pi/6}^{\pi/3}dx$

57. Putting t = 1/u in I_2 we have

$I_2=-\int_e^{\tan x}\frac{\mu d\mu}{1+\mu^2}=-\int_{1/e}^{\tan x}\frac{\mu d\mu}{1+\mu^2}+\int_{1/e}^{e}\frac{\mu d\mu}{1+\mu^2}$

$=-I_1+\frac{1}{2}\int_{1/e}^{e}\frac{2\mu d\mu}{1+\mu^2}$ so

$I_1+I_2=\frac{1}{2}\log\left(\mu^2+1\right)]^e 1/e=\frac{1}{2}\left[\log\left(e^2+1\right)-\log\left(\frac{e^2+1}{e^2}\right)\right]$

$\frac{1}{2}\times 2=1$

58. $I=\int_0^{\pi}\frac{(\pi-x)(\sin 2x)\left(-\sin\left(\frac{\pi}{2}\cos x\right)\right)}{\pi-2x}dx$

then $2I=\int_0^{\pi}\left(\sin 2x.\sin\left(\frac{\pi}{2}\cos x\right)\right)dx$

$=4\int_0^{\pi/2}\cos x\sin\left(\frac{\pi}{2}\cos x\right)\sin x\,dx$

$\Rightarrow I=2\int\left(\frac{2}{\pi}.t\right)\sin t\left(\frac{2}{\pi}.dt\right)$

(where $t=\frac{\pi}{2}\cos x$) $\Rightarrow I=\frac{8}{\pi^2}\int_0^{\pi/2}t\sin t\,dt$

$=\frac{8}{\pi^2}(-t\cos t+\sin t)_0^{\pi/2}=\frac{8}{\pi^2}$

59. $2A+B=8,\frac{A}{3}+\frac{B}{2}=\frac{8}{3}$; $f^{'}(x)=1$

60. $f(x)=ae^{2x}+be^{x}+cx$

$f'(x)=2ae^{2x}+be^{x}+c$

$f(0)=-1\Rightarrow a+b=-1$ ----------(1)

$f'(\log 2)=31\Rightarrow 8a+2b+c=31$ -----(2)

$\int_0^{\log 4}(f(x)-cx)dx=\frac{39}{2}$

LEVEL-IV

STATEMENT TYPE QUESTIONS

1) Statement-I is True, Statement-II is True; Statement-II is a correct explanation for Statement-I.
2) Statement-I is True, Statement-II is True; Statement-II is NOT a correct explanation for Statement-I.
3) Statement-I is True, Statement-II is False.
4) Statement-I is False, Statement-II is True.

1. **Statement-I:** $\int_0^{\pi/2}\frac{dx}{1+\tan^5 x}=\frac{\pi}{4}$

Statement-II: $\int_0^a f(x)dx=\int_0^a f(a-x)dx,$

2. **Statement-I:**

$\int_0^{\pi}x\sin x\cos^2 x\,dx=\frac{\pi}{2}\int_0^{\pi}\sin x\cos^2 x\,dx$.

Statement-II: $\int_a^b x\,f(x)dx=\frac{a+b}{2}\int_a^b f(x)dx$.

3. **Statement I:** $\int_0^{\pi/2}\frac{5\tan x-3\cot x}{\tan x+\cot x}dx=\frac{\pi}{2}$.

Statement II:

$\int_0^{\pi/2}\frac{a\tan x+b\cot x}{\tan x+\cot x}dx=(a+b)\frac{\pi}{4}$.

4. **Statement-I:** The value of

$\int_0^1\tan^{-1}\frac{2x-1}{(1+x-x^2)}dx=0$.

Statement-II: $\int_a^b f(x)dx=\int_a^b f(a+b-x)dx$.

5. **Statement-I:** $\int_0^{\pi}\sqrt{1-\sin^2 x}\,dx=0$.

Statement-II: $\int_0^{\pi}\cos x\,dx=0$.

6. **Statement-I:** If f statisfies

$f(x+y)=f(x)+f(y)\forall x,y\in R$ then $\int_{-5}^{5}f(x)dx=0$.

Statement-II: If f is an odd function then $\int_{-a}^{a}f(x)dx=0$

7. **Statement I:** $\int_0^{\pi/2}\sin^7 x\,dx=\frac{16}{35}$.

Statement II:

$\int_0^{\pi/2}\sin^n x\,dx=\frac{n-1}{n}.\frac{n-3}{n-2}.\frac{n-5}{n-4}.....\frac{2}{3}.1$ if n is odd.

8. **Statement-I:** The value of $\int_0^{\pi}\sin^{100}x\cos^{99}x\,dx$ is zero

Statement-II: $\int_a^b f(x)dx=\int_{a+c}^{b+c}f(x-c)dx$.

9. **Statement-I:** If $I_a = \int_a^x \tan t\, dt$, then $5(I_4 + I_6) = \tan^5 x + C$ (where C is real constant).

Statement-II: If $I_n = \int_a^x \tan^4 t\, dt$, then $\frac{\tan^{n-1} x}{n} - I_{n-2} = I_n, n \in N$.

10. **Statement-I:** There exist a number $x \in [a,b]$ such that $\int_a^x f(t)\, dt = \int_x^b f(t)\, dt$.

Statement-II: $f(x)$ is a continuous function.

11. **Statement-I:** $\int_0^{4\pi} [|\sin x| + |\cos x|]\, dx$, (where [.] denotes G.I.F.) equals 8π.

Statement-II: If $f(x) = |\sin x| + |\cos x|$, then $1 \le f(x) \le \sqrt{2}$.

12. **Statement-I:** $f(x) = \int_1^x \frac{\log t\, dt}{1+t+t^2} (x>0)$ then $f(x) = -f\left(\frac{1}{x}\right)$.

Statement-II: If $f(x) = \int_1^x \frac{\log t\, dt}{t+1}$ then $f(x) + f\left(\frac{1}{x}\right) = \frac{1}{2}(\log x)^2$.

13. **Let $f(x)$ is continuous and positive for $x \in [a,b], g(x)$ is continuous for $x \in [a,b]$ and $\int_a^b |g(x)|\, dx > \left|\int_a^b g(x)\, dx\right|$, then**

Statement-I: The value of $\int_a^b f(x) g(x)\, dx$ can be zero.

Statement-II: Equation $g(x) = 0$ has at least one root in $x \in (a,b)$.

14. **Consider the funcion $f(x)$ satisfying the relation $f(x+1) + f(x+7) = 0, \forall x \in R$.**

Statement-I: The possible least value of 't' for which $\int_a^{a+t} f(x)\, dx$ is indipendent of 'a' is 12.

Statement-II: $f(x)$ is independent of a is 12.

15. **Consider $I_1 = \int_0^{\pi/4} e^{x^2} dx, I_2 = \int_0^{\pi/4} e^x dx$, $I_3 = \int_0^{\pi/4} e^{x^2} \cos x\, dx, I_4 = \int_0^{\pi/4} e^{x^2} \sin x\, dx$.**

Statement-I: $I_2 > I_1 > I_3 > I_4$.

Statement-II: For $x \in (0,1), x > x^2$ and $\sin x > \cos x$.

16. **Statement-I:** If $f(x)$ is continuous on $[a,b]$, then there exists a point $c \in (a,b)$ such that $\int_a^b f(x)\, dx = f(c)(b-a)$.

Statement-II: For $a < b$, if m and M are, respectively, the smallest and greatest values of $f(x)$ on $[a,b]$, then

$$m(b-a) \le \int_a^b f(x)\, dx \le (b-a)M.$$

17. **Statement-I:** If y is a function of x such that $y(x-y)^2 = x$ then $\int \frac{dx}{x-3y} = \frac{1}{2}\left[\log(x-y)^2 - 1\right]$.

Statement-II: $\int \frac{dx}{x-3y} = \log(x-3y) + c$

18. **Statement-I:** Let m be any integer. Then the value of $I_m = \int_0^{\pi} \frac{\sin 2mx}{\sin x} dx$ is zero.

Statement-II: $I_1 = I_2 = I_3 = ...I_m$.

19. **Statement-I:** The value of $\int_0^{\pi/4} \log(1 + \tan\theta)\, d\theta = \frac{\pi}{8} \log 2$.

Statement-II: The value of $\int_0^{\pi/2} \log \sin\theta\, d\theta = -\pi \log 2$.

20. **Statement I:** $\int_0^{\pi/2} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx = \frac{\pi}{4}$.

Statement II: $\int_0^{\pi/2} \frac{f(\sin x)}{f(\sin x) + f(\cos x)} dx = \frac{\pi}{4}$

21. **Statement-I:** If $n > 1$ then $\int_0^{\infty} \frac{dx}{1+x^n} = \int_0^1 \frac{dx}{(1-x^n)^{1/n}}$

Statement-II: $\int_a^b f(x)\, dx = \int_a^b f(a+b-x)\, dx$.

22. **Statement-I:** The value of $\int_0^{2\pi} \cos^{99} x\,dx$ is 0.

Statement-II: $\int_0^{2a} f(x)\,dx = 2\int_0^{a} f(x)\,dx$, if $f(2a-x) = f(x)$.

23. **Statement-I:** $\int_{-1/2}^{1/2} \log\left(\frac{1+x}{1-x}\right)dx = 0$

Statement-II: If 'f' is an odd function then $\int_{-a}^{a} f(x)\,dx = 0$

24. **Statement-I:** $\int_a^x f(t)\,dt$ is an even function if $f(x)$ is an odd function.

Statement-II: $\int_a^x f(t)\,dt$ is an odd function if $f(x)$ is an even function.

25. **Statement I:** $\int_{-\pi/2}^{\pi/2} \sin x.\cosh x\,dx = 0$

Statement II: If $f(x)$ is an odd function then $\int_{-a}^{a} f(x)\,dx = 0$.

26. **Statement I:** $\int_0^{100\pi} \sqrt{1-\cos 2x}\,dx = 200\sqrt{2}$.

Statement II: If $f(x)$ is a periodic function with period a then $\int_0^{na} f(x)\,dx = n\int_0^{a} f(x)\,dx$.

27. **Statement I:** $\int_0^{\pi/4} \left(\tan^6 x + \tan^4 x\right)dx = \frac{1}{5}$.

Statement II: $\int_0^{\pi/4} \left(\tan^n x + \tan^{n-2} x\right)dx = \frac{1}{n-1}$

28. **Statement I:** $\int_0^{\pi/4} \tan^5 x\,dx = \frac{1}{2}\log 2 - \frac{1}{4}$.

Statement II: If $I_n = \int_0^{\pi/4} \tan^n x\,dx$ then $I_n = \frac{1}{n-1} - \frac{1}{n-3} + \frac{1}{n-5} - .....I_0$ or I_1 according as n is even or odd where $I_0 = \frac{\pi}{4}, I_1 = \frac{1}{2}\log 2$.

29. Statement-I: $\int_0^x |\sin t|\,dt$, for $x \in [0, 2\pi]$ is a non-differentiable function.

Statement-II: $|\sin t|$ is non-differentiable at $x = \pi$.

30. **Statement-I:** The value of $\int_{-4}^{-5} \sin\left(x^2 - 3\right)dx + \int_{-2}^{-1} \sin\left(x^2 + 12x + 33\right)$ is zero.

Statement-II: $\int_{-a}^{a} f(x)\,dx = 0$ if $f(x)$ is an odd function.

31. **Statement-I:** On the interval $\left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$, the least value of the function $f(x) = \int_{5\pi/4}^{x} (3\sin t + 4\cos t)\,dt$ is zero.

Statement-II: If $f(x)$ is a decreasing function on the interval $[a, b]$, then the least value of $f(x)$ is $f(b)$.

32. **Statement-I:** $f(x)$ is symmetrical about $x = 2$, then $\int_{2-a}^{2+a} f(x)\,dx$ is equal to $2\int_2^{2+a} f(x)\,dx$.

Statement-II: If $f(x)$ is symmetrical about $x = b$, then $f(b-\alpha) = f(b+\alpha)\,\forall(\alpha \in R)$.

33. **Statement-I:** If $\int_0^{\pi/2} \sin^4 x\,dx = k$ then $\int_{-\pi}^{\pi} \sin^4 x\,dx = 8k$

Statement-II : Given $\int_0^1 \frac{x}{\sqrt{1-x^2}}\,dx = 1$ then

$\underset{n\to\infty}{Lt} \sum_{r=1}^{n} \sqrt{\frac{n+r}{n^2(n-r)}} = \frac{\pi}{2} + 1$

which of the above statement is true

1) Only I
2) Only II
3) Both I and II
4) Neither I nor II

ASSERTION AND REASON TYPE QUESTIONS

1) Both A and R are individually true and R is the correct explanation of A
2) Both A and R are individually true but R is not the correct explanation of A
3) A is true but R is false.
4) A is false but R is true

34. **Assertion (A):**
$$\int_{\pi/6}^{\pi/3}\frac{(\sin x)^{\sqrt{2}}}{(\sin x)^{\sqrt{2}}+(\cos x)^{\sqrt{2}}}dx=\frac{\pi}{12}.$$
Reason (R): $\int_{\pi/6}^{\pi/3}\frac{f(x)dx}{f(x)+f\left(\frac{\pi}{2}-x\right)}dx=\frac{\pi}{12}$.

35. **Assertion (A):** $f(x)=\int_0^x \ln\left(\frac{1-t}{1+t}\right)dt$ **is an even function.**
Reason (R): $\ln\left(\frac{1-t}{1+t}\right)$ **is an odd function.**

36. **Assertion (A): If** $\int_0^1 e^{\sin x}dx=\lambda$ **, then** $\int_0^{200}e^{\sin x}dx=200\lambda$.
Reason (R): $\int_0^{na}f(x)dx=n\int_0^a f(x)dx,\ n\in I$ **and** $f(a+x)=f(x)$.

37. **Assertion (A):** $\int_0^{\pi/2}\left(\sin^6 x+\cos^6 x\right)dx$ lies in the interval $\left(\frac{\pi}{8},\frac{\pi}{2}\right)$.
Reason (R): $\sin^6 x+\cos^6 x$ is periodic with period $\pi/2$.

38. **Assertion (A):** $\int_{-\pi/2}^{\pi/2}|\sin x|dx=2$.
Reason (R):
$\int_a^b f(x)dx=\int_a^c f(x)dx+\int_c^b f(x)dx$, **where** $c\in(a,b)$

39. **Assertion (A):**
$$\lim_{n\to\infty}\left(\frac{1}{n+1}+\frac{1}{n+2}+\frac{1}{n+3}+...+\frac{1}{2n}\right)=\ln 3.$$
Reason (R): $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{n}f\left(\frac{r}{n}\right)=\int_0^1 f(x)dx$.

40. **Assertion (A):** $\int_{\pi/2}^{3\pi/2}[2\sin x]dx=0$, where $[.]$ denotes the greatest integer function.
Reason (R): $2\sin x$ decreasing function in $\left(\frac{\pi}{2},\frac{3\pi}{2}\right)$.

41. **Assertion (A):** $\int_0^5[x]dx=10$
Reason (R): $\int_a^b\left|x^2-(a+b)x+ab\right|dx=\frac{(b-a)^3}{6}$

42. **Assertion :** $\int_0^{\pi/2}\frac{\sin x}{\sin x+\cos x}dx=\frac{\pi}{4}$
Reason : $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

43. **Assertion (A):** $\int_0^{1/4}\frac{dx}{1+(\cot 2\pi x)^{\sqrt{2}}}=\frac{1}{4}$.
Reason (R): If $a+b=\pi/2$, then $\int_a^b\frac{f(\sin x)}{f(\sin x)+f(\cos x)}dx=\frac{(b-a)}{2}$.

44. **Assertion :** $\int_0^{100\pi}|\sin x|dx=200$
Reason : $|\sin x|$ is a periodic function of period 2π and $\int_0^{\pi}|\sin x|dx$ is 4

45. **Assertion (A):** $\int_0^{2\pi}\sin^3 x\,dx=0$.
Reason (R): $\sin^3 x$ is an odd function.

46. **Assertion (A):** $\int_{-1}^{1}\frac{|x|}{x}dx=2$.
Reason (R): $\frac{|x|}{x}$ is undefined at $x=0$.

47. **Assertion (A):** $\int_0^t\cos x\,dx=\sin t$.
Reason (R): $\cos x$ is continuous in any closed interval $[0,t]$.

48. **Assertion (A):** $\int_1^6\{x+5\}^2dx=41$, where $\{.\}$ denotes the fractional part function.
Reason (R): $\{x+5\}$ is a periodic function.

49. Assertion (A): $\int_0^2 f(x)\,dx = \frac{4(\sqrt{2}-1)}{3}$, where

$$f(x)=\begin{cases} x^2, & \text{for } 0\le x<1 \\ \sqrt{x} & \text{for } 1\le x\le 2\end{cases}.$$

Reason (R): $f(x)$ is continuous in $[0,2]$.

50. Assertion (A):

$\int_0^1 e^{-x}\cos^2 x\,dx < \int_0^1 e^{-x^2}\cos^2 x\,dx$.

Reason (R):

$\int_a^b f(x)\,dx < \int_a^b g(x)\,dx \ \forall f(x)\ge g(x)$.

51. Assertion (A): $\int_0^{\pi/2}\frac{\sin x}{x}dx < \frac{\pi}{2}$.

Reason (R): $\underset{x\to 0}{Lt}\frac{\sin x}{x}=1$.

52. Assertion (A): If $f(x)=x-x^2+1$ and $g(x)=\max\{f(t):0\le t\le x\}$, then $\int_0^1 g(x)\,dx=\frac{29}{24}$.

Reason (R): $f(x)$ is increasing in (0, 1/2) and decreasing in (1/2, 1).

MATRIX MATCHING TYPE

53. Observe the following Lists

List-I	List-II
A) $\int_0^{\pi/2}\log\sin x\,dx =$	1. $-\frac{\pi^2}{2}\log 2$
B) $\int_0^{\pi/2}\log\tan x\,dx =$	2. $\frac{-\pi}{2}\log 2$
C) $\int_0^{\pi} x\log\sin x\,dx =$	3. 4π
D) $\int_{-\pi}^{\pi}\left(x^3+x\cos x+\tan^5 x+2\right)dx$	4. 0

The correct Match for List-I from List-II is

	A	B	C	D		A	B	C	D
1)	3	4	1	2	2)	1	4	2	3
3)	4	3	2	1	4)	2	4	1	3

54. Observe the following Lists

List-I	List-II
A) $\int_0^a f(x)\,dx$	1) $\int_0^a f(a+x)\,dx$
B) $\int_{-a}^a f(x)\,dx$	2) $\int_0^a f(a-x)\,dx$
C) $\int_0^{2a} f(x)\,dx$	3). 0, if f(x) is odd
D) $\int_0^{na} f(x)\,dx$	4).0, if f (2a-x) =- f(x)
	5) $n\int_0^a f(x)\,dx$, If period of f(x) is a
	6)(n-1) $\int_0^a f(x)\,dx$, if period of f(x) is a

The correct match for List-I from List-II

	A	B	C	D
1)	1	2	4	6
2)	2	3	4	5
3)	2	3	4	6
4)	2	1	4	5

55. Observe the following Lists

List-I	List-II
A: $\int_{-2}^{2}\frac{1}{4+x^2}dx$	1) $\frac{\pi}{3}$
B: $\int_1^2\frac{1}{x\sqrt{x^2-1}}dx$	2) 0
C: $\int_0^{\pi}\cos 3x.\cos 2x\,dx$	3) $\frac{\pi}{4}$
	4) $\frac{\pi}{2}$

The correct match for List-I from List-II

	A	B	C
1)	3	1	4
2)	3	1	2
3)	1	3	2
4)	4	1	2

56. Observe the following Lists

List-I	List-II
A) $\int_0^{\infty} e^{-4x}\sin 5x\,dx$	P. 3
B) $\int_2^8 \frac{[x^2]dx}{[x^2-20x+100]+[x^2]}$ (where [] is GIF	Q. $\frac{5}{41}$
C) $\int_0^{\pi/2}\left[x^n+n(n-1)x^{n-2}\cos x\right]dx$ (where $n\varepsilon N$)	R. 120
D) $\int_0^{\infty} x^5 e^{-x}dx$	S. $\left(\frac{\pi}{2}\right)^n$

1) A - Q, B - P, C - S, D - R
2) A - P, B-Q, C- R, D- S
3) A -S, B-Q, C-R, D-P
4) A -R, B-P, C-S, D-Q

57. Observe the following Lists

List-I	List-II
A) $\int_0^{\pi/2} e^{\sin^2 x}\sin 2x\ \ dx$	p. $\frac{\pi}{4}$
B) $\int_1^1 x\lvert x\rvert dx$	q. e -1
C) $\int_0^{\pi/2}\frac{(\sin x)^{5/2}}{(\sin x)^{5/2}+(\cos x)^{5/2}}dx$	r. $\frac{\pi}{32}$
D) $\int_0^{\pi/2}\sin^4 x\cos^2 x dx$	s. 0

1) A -q, B- s, C- p, D-r
2) A -s, B- p, C- q, D-r
3) A -p, B- s, C-q, D-r
4) A -s, B- r, C -p, D-q

INCREASING & DECREASING

58. If $I_1=\int_0^{\pi/2}\sin^4 x dx$; $I_2=\int_0^{\pi/2}\cos^6 x dx$

$I_3=\int_0^{\pi/2}\sin^8 x dx$ $I_4=\int_0^{\pi/2}\cos^2 x dx$

then the increasing order of I_1,I_2,I_3,I_4 is

1) I_4,I_2,I_3,I_1 2) I_3,I_2,I_1,I_4
3) I_4,I_3,I_2,I_1 4) I_3,I_2,I_4,I_1

59. The values of the following definite integrals in the decreasing order is

A) $\int_0^3 [x]dx$ B) $\int_{-2}^2 x^3 dx$

C) $\int_0^3 (x-[x])dx$ D) $\int_{-1}^1 \lvert x\rvert dx$

1) A,C,D,B 2) A,D,C,B
3) A,D,C,B 4) A,C,B,D

LEVEL-IV KEY

1) 1	2) 3	3) 1	4) 1	5) 4	6) 1
7) 1	8) 1	9) 3	10) 1	11) 4	12) 4
13) 1	14) 1	15) 3	16) 1	17) 3	18) 1
19) 3	20) 1	21) 2	22) 1	23) 1	24) 3
25) 1	26) 1	27) 1	28) 1	29) 4	30) 2
31) 4	32) 1	33) 2	34) 1	35) 1	36) 4
37) 2	38) 2	39) 4	40) 4	41) 2	42) 1
43) 4	44) 3	45) 2	46) 4	47) 1	48) 4
49) 4	50) 3	51) 2	52) 1	53) 4	54) 2
55) 2	56) 1	57) 1	58) 2	59) 1	

LEVEL-IV HINTS

1. (1) is correct $\int_0^a f(x)dx=\int_0^a f(a-x)dx$

$$I=\int_0^{x/2}\frac{dx}{1+\tan^5 x}=\int_0^{x/2}\frac{dx}{1+\tan^5\left(\frac{\pi}{2}-x\right)}$$

$$I=\int_0^{\pi/2}\frac{\tan^5 x\,dx}{1+\tan^5 x}$$

$$I+1=\int_0^{\pi/2}\frac{1+\tan^5 x}{1+\tan^5 x}dx=\int_0^{x/2}1dx=\frac{\pi}{2}\quad I=\frac{\pi}{4}$$

2. $\int_a^b x f(x)dx=\int_a^b (a+b-x)f(a+b-x)dx$

$$=(a+b)\int_a^b f(a+b-x)dx-\int_a^b f(a+b-x)dx$$

$\therefore$ Statement II is true only when $f(a+b-x)=f(x)$ which holds in statement I.

$\therefore$ Statement II is false and statement I is true

3. $\int_0^{\pi/2} \frac{a\tan x + b\cot x}{\tan x + \cot x} dx = (a+b)\frac{\pi}{4}$

4. $I = \int_0^1 \tan^{-1} \frac{2(1-x)-1}{1+(1-x)-(1-x)^2} dx$

$= \int_0^1 \tan^{-1} \frac{1-2x}{1+x-x^2} dx \quad = -I \quad \Rightarrow I = 0$

5. $\int_0^\pi \sqrt{1-\sin^2 x}\, dx \quad = \int_0^\pi |\cos x| dx$

$= \int_0^{\pi/2} \cos x\, dx + \int_{\pi/2}^{\pi} -\cos x\, dx \qquad = 1+1 = 2$

Hence, statement 1 is false.
However, statement 2 is true.

6. (1) is correct as $f(x) = Kx$

$f(x+y) = f(x) + f(y) \forall x, y$

$\Rightarrow f(x) = Kx$ - $\int_{-5}^{5} f(x) dx = \int_{-5}^{5} Kx\, dx = 0$

7. $\int_0^{\pi/2} \sin^n x\, dx = \frac{n-1}{n} \times \frac{n-3}{n-2} \times ..... \times \frac{2}{3} \times 1$

8. To prove $\int_a^b f(x) dx = \int_{a+c}^{b+c} f(x-c) dx$

Put $z = x - c$, then $dz = dx$

When $x = a+c, z = a$ and when $x = b+c, z = b$

$\therefore \int_{a+c}^{b+c} f(x-c) dx = \int_a^b f(z) dz = \int_a^b f(x) dx$

Thus, statement II is true

$\int_a^b f(x) dx = \int_{a+c}^{b+c} f(x-c) dx$

Putting $f(x) = \sin^{100} x \cos^{99} x, a = 0, b = \pi$, and

$c = -\frac{\pi}{2}$, we get $\int_0^\pi \sin^{100} x \cos^{99} x\, dx$

$= \int_{-\pi/2}^{\pi/2} \sin^{100}\left(x + \frac{\pi}{2}\right) \cos^{99}\left(x + \frac{\pi}{2}\right) dx$

$= -\int_{-\pi/2}^{\pi/2} \cos^{100} x \sin^{99} x\, dx$

$= 0 \left[\because \cos^{100} x \sin^{99} x \text{ is an odd function}\right]$

9. (3) is correct

10. Let $g(x) = \int_a^x f(t) dt - \int_x^b f(t) dt$, where

$x \in [a,b]$ We have $g(a) = -\int_a^b f(t) dt$ and

$g(b) = \int_a^b f(t) dt$

$\Rightarrow g(a) g(b) = -\left(\int_a^b f(t) dt\right)^2 \le 0$

Clearly, $g(x)$ is continuous in $[a,b]$ and $g(a) g(b) \le 0$. It implies that $g(x)$ will becomes zero at least once in $[a,b]$. Hence, $\int_a^x f(t) dt = \int_x^b f(t) dt$ for all atleast one value of $x \in [a,b]$.

Hence, both the statements are true and statement II is a correct explanation of statement I.

11. (4) is correct $1 \le |\sin x| + |\cos x| \le \sqrt{2}$

$\Rightarrow [|\sin x| + |\cos x|] = 1$

$I = \int_0^{4\pi} [|\sin x| + |\cos x|] dx = \int_0^{4\pi} 1.dx = 4\pi$

12. $f\left(\frac{1}{x}\right) = \int_1^{1/x} \frac{\log t\, dt}{1+t+t^2}$

on putting $t = \frac{1}{z}, dt = -\frac{1}{z^2} dz$;

$f\left(\frac{1}{x}\right) = \int \frac{\log 1/z}{1 + \frac{1}{z} + \frac{1}{z^2}} \left(-\frac{dz}{z^2}\right)$

$= \int_1^x \frac{\log z\, dz}{z^2 + z + 1} = \int_1^x \frac{\log t\, dt}{1+t+t^2} = f(x)$

$\Rightarrow$ Assertion A is false The reason R is true which can be proved in the same style.

13. Given that $\int_a^b |g(x)| dx > \left|\int_a^b g(x) dx\right| \Rightarrow y = g(x)$ cuts the graph at least once, then $y = f(x) g(x)$

changes sign at least once in (a,b), hence

$\int_a^b f(x)g(x)dx$ can be zero

14. Given $f(x+1)+f(x+7)=0, \forall x \in R$

Replace x by $x-1$, we have

$f(x)+f(x+6)=0$ ..... (1)

Now replace x by $x+6$, we have

$f(x+6)+f(x+12)=0$ ..... (2)

From equations (1) and (2), we have

$f(x)=f(x+12)$ ..... (3)

Hence, $f(x)$ is periodic with period 12.

$\Rightarrow \int_a^{a+t} f(x)dx$ is independent of a if t is positive integral multiple of 12 then possible value of t is 12.

15. $x > x^2 \forall x \in \left(0,\frac{\pi}{4}\right) \Rightarrow e^x > e^{x^2} \forall x \in \left(0,\frac{\pi}{4}\right)$

$\cos x > \sin x \forall x \in \left(0,\frac{\pi}{4}\right)$

$\Rightarrow e^{x^2}\cos x > e^{x^2}\sin x$

$\Rightarrow e^x > e^{x^2} > e^{x^2}\cos x > e^{x^2}\sin x \forall x \in \left(0,\frac{\pi}{4}\right)$

$\Rightarrow I_2 > I_1 > I_3 > I_4$

16. For $a<b$. If m and M are the smallest and greatest values of $f(x)$ on $[a,b]$

then $m(b-a) \le \int_a^b f(x)dx \le (b-a)M$

or $m \le \frac{1}{(b-a)}\int_a^b f(x)dx \le M$

Since $f(x)$ is continuous on $[a,b]$, it takes on all intermediate values between m and M.

Therefore, some values $f(c)(a \le f(c) \le b)$, we will have $\frac{1}{(b-c)}\int_a^b f(x)dx = f(c)$ or

$\int_a^b f(x)dx = f(c)(b-a)$

Hence, both the statements are true and statement II is a correct explanation of statement I.

17. The Statement-II is false since in writing $\int \frac{dx}{x-3y} = \log(x-3y)+C$, we are assuming that is a constant. We will now prove the assertion.

From the given relation $(x-y)^2 = \frac{x}{y}$ and

$2\log(x-y) = \log x - \log y$

Also $\frac{dy}{dx} = \left(-\frac{y}{x}\right).\frac{x+y}{x-3y}$. To prove the integral relation it is sufficient to show that

$\frac{d}{dx}RHS = \frac{1}{x-3y}$

Now $= \frac{1}{2}\log\left[\frac{x}{y}-1\right] \left(\because (x-y)^2 = \frac{x}{y}\right)$

$= \frac{1}{2}\left[\log(x-y) - \log y\right]$

$= \frac{1}{2}\left[\frac{\log x - \log y}{2} - \log y\right]$

$= \frac{1}{4}\left[\log x - 3\log y\right]$

$\Rightarrow \frac{d}{dx}RHS = \frac{1}{4}\left[\frac{1}{x} - \frac{3}{y}\frac{dy}{dx}\right]$

$= \frac{1}{4}\left[\frac{1}{x} - \frac{3}{y}\left(-\frac{y}{x}\right)\frac{x+y}{x-3y}\right] = \frac{1}{x-3y}$

18. Let $I_m = \int_0^\pi \frac{\sin 2mx}{\sin x}dx$. then,

$I_m - I_{m-1} = \int_0^\pi \frac{\sin 2mx - \sin 2(m-1)x}{\sin x}dx$

$= \int_0^\pi 2\cos(2m-1)x\,dx$

$=\frac{2}{2m-1}\left[\sin(2m-1)x\right]_0^\pi=0$

$I_m=I_{m-1}$ for all $m\in N$

$\Rightarrow I_m=I_{m-1}=I_{m-2}=...=I_1$

But, $I_1=\int_0^\pi\frac{\sin 2x}{\sin x}dx=2\int_0^\pi\cos x\,dx=0$

$\therefore I_m=0$ for all $m\in N$

19. Both the statements are true independently, but statement II is not a correct explanation of statement I.

20. $\int\limits_0^{\pi/2}\frac{f(\sin x)}{f(\sin x)+f(\cos x)}dx=\frac{\pi}{4}$

21. The Assertion A can be be proved by showing that both integrals are equal to a third integral. Put $x^n=\tan^2\theta$ in the integral on LHS and $x^n=\sin^2\theta$ in the integral on RHS, then both integrals will be equal to $\frac{2}{\pi}\int_0^{\pi/2}\tan^{\frac{2}{n}-1}\phi d\phi$ and $\frac{2}{\pi}\int_0^{\pi/2}\tan^{\frac{2}{n}-1}\phi d\phi$ respectively.

Since the last two integrals are equal assertion is proved. the correct reason R has no role to play here. $\Rightarrow$ Choice (2) is correct

22. Let $I=\int_0^{2\pi}\cos^{99}x\,dx$ Then,

$I=2\int_0^\pi\cos^{99}x\,dx\ \left[\because\cos^{99}(2\pi-x)=\cos^{99}x\right]$

Now,

$I=\int_0^\pi\cos^{99}x\,dx=0\ \left[\because\cos^{99}(\pi-x)=-\cos^{99}x\right]$

$\Rightarrow I=2\times0=0$

23. $f(x)=\log\left(\frac{1+x}{1-x}\right)$; $f(-x)=\log\left(\frac{1-x}{1+x}\right)$

$-\log\left(\frac{1+x}{1-x}\right)\qquad f(-x)=-f(x)$

so $f(x)$ is an odd function $\Rightarrow\int\limits_{-a}^{a}f(x)dx=0$

24. Statement I is true as it is a fundamental property. (See integration of odd and even function.)

Let $g(x)=\int\limits_a^x f(t)dt$ If $f(x)$ is an even function

Then $g(-x)=\int\limits_a^{-x}f(t)dt=-\int\limits_{-a}^{x}f(-y)dy$

$=-\int\limits_{-a}^{x}f(y)dy=-\int\limits_{-a}^{a}f(y)dy-\int\limits_a^x f(y)dy\neq-g(x)$

Hence, statement II is false.

25. $f(x)$ is odd.

26. If $f(x)$ is a periodic function with period "a" then $\int_0^{na}f(x)dx=n\int_0^a f(x)dx$.

27. $\int\limits_0^{\pi/4}\left(\tan^n x+\tan^{n-2}x\right)dx=\frac{1}{n-1}$

28. If $I_n=\int\limits_0^{\pi/4}\tan^n x\,dx\,(n>1)$ then $I_n+I_{n-2}=\frac{1}{n-1}$

29. Obviously, $|\sin t|$ is non-differentiable at $x=\pi$ But

$$\int_0^x|\sin t|dt=\begin{cases}\int_0^x\sin t\,dt, & 0\le x<\pi\\ \int_0^\pi\sin t\,dt+\int_\pi^x-\sin t\,dt, & \pi\le x\le 2\pi\end{cases}$$

$$=\begin{cases}-\cos x+1, & 0\le x<\pi\\ 3+\cos x, & \pi\le x\le 2\pi\end{cases}$$

Which is continuous as well as differentiable at $x=\pi$. Hence, statement I is false

30. $I=\int_{-4}^{-5}\sin(x^2-3)dx$

$+\int_{-2}^{-1}\sin(x^2+12x+33)dx=I_1+I_2$

$I_2=\int_{-2}^{-1}\sin(x^2+12x+33)dx$

$=\int_{-2}^{-1}\sin\left((x+6)^2-3\right)dx$, put $x+6=-y$

$\Rightarrow I_2=-\int_{-4}^{-5}\sin(y^2-3)dy=-I_1$

$\Rightarrow I_1+I_2=0\Rightarrow I=0$

31. $f(x)=\int_{5\pi/4}^{x}(3\sin t+4\cos t)dt$

$\Rightarrow f'(x) = 3\sin x + 4\cos x, x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

These values of x are in third quadrant where both $\sin x$ and $\cos x$ are negative.

Then $f'(x) < 0$ for $x \in \left[\frac{5\pi}{4}, \frac{4\pi}{3}\right]$

Hence, $f(x)$ is decreasing for these values of x

Then, the least value of function occurs at $x = \frac{4\pi}{3}$

$\Rightarrow f_{\text{minimum}} = \int_{5\pi/4}^{4\pi/3} (3\sin t + 4\cos t)\,dt = \frac{3}{2} + \frac{1}{\sqrt{2}} - 2\sqrt{3}$

32. Statement II is a fundamental concept, also we have $f(2-a) = f(2+a)$

$$\int_{2-a}^{2+a} f(x)\,dx = 2\int_{2}^{2+a} f(x)\,dx$$

33. (1) $\sin^4 x$ is an even function

(2) Given $= \underset{n\to\infty}{Lt} \frac{1}{n}\sum_{r=1}^{n} \sqrt{\frac{1+\frac{r}{n}}{1-\frac{r}{n}}}$

34. $I = \int_{\pi/6}^{\pi/3} \frac{f(x)\,dx}{f(x) + f\left(\frac{\pi}{2} - x\right)}$ ...... (1)

$$= \int_{\pi/6}^{\pi/3} \frac{f\left(\frac{\pi}{3} + \frac{\pi}{6} - x\right)dx}{f\left(\frac{\pi}{3} + \frac{\pi}{6} - x\right) + f\left(\frac{\pi}{2} - \left(\frac{\pi}{3} + \frac{\pi}{6} - x\right)\right)}$$

(by property)

$$= \int_{\pi/6}^{\pi/3} \frac{f\left(\frac{\pi}{2} - x\right)dx}{f\left(\frac{\pi}{2} - x\right) + f(x)} \text{ ...... (2)}$$

Adding Equation (1) and (2), then

$2I = \int_{\pi/6}^{\pi/3} 1.dx = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$ $\quad \therefore I = \frac{\pi}{12}$

Also, for $f(x) = (\sin x)^{\sqrt{2}}, I = \frac{\pi}{12}$

35. $\because f(x) = \int_0^x \ln\left(\frac{1-t}{1+t}\right)dt \quad \because f'(x) = \ln\left(\frac{1-x}{1+x}\right)$

$\Rightarrow f'(-x) = \ln\left(\frac{1+x}{1-x}\right) = -\ln\left(\frac{1-x}{1+x}\right) = -f'(x)$

$\Rightarrow f'(x)$ is an odd function.

Hence, $f(x)$ is an even function

36. $\because$ Period of $e^{\sin x}$ is 2π $\quad \therefore \int_0^{200} e^{\sin x}dx \neq 200\lambda$

37. $\therefore \sin^6 x + \cos^6 x = (\sin^2 x)^3 + (\cos^2 x)^3$

$= (\sin^2 x + \cos^2 x)^3 - 3\sin^2 x\cos^2 x(\sin^2 x + \cos^2 x)$

$= 1 - 3\sin^2 x\cos^2 x = 1 - \frac{3}{4}\sin^2 2x \left(\therefore \text{period}\ \frac{\pi}{2}\right)$

$\therefore$ Least and greates value of $\sin^6 x + \cos^6 x$ are $\frac{1}{4}$ and 1 Hence,

$\left(\frac{\pi}{2} - 0\right) \times \frac{1}{4} < \int_0^{\pi/2} (\sin^6 x + \cos^6 x)\,dx < \left(\frac{\pi}{2} - 0\right) \times 1$

$\Rightarrow \frac{\pi}{8} < \int_0^{\pi/2} (\sin^6 x + \cos^6 x)\,dx < \frac{\pi}{2}$

38. $\because |\sin x|$ is an even function

$\therefore \int_{-\pi/2}^{\pi/2} |\sin x|\,dx = 2\int_0^{\pi/2} |\sin x|\,dx = 2\int_0^{\pi/2} \sin x\,dx$

$= -2(\cos x)_0^{\pi/2} = -2(0-1) = 2$

39. $\because \lim_{n\to\infty}\left(\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \ldots + \frac{1}{2n}\right)$

$= \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{(n+r)} = \lim_{n\to\infty} \sum_{r=1}^{n} \frac{1}{n} \cdot \frac{1}{\left(1 + \frac{r}{n}\right)}$

$= \int_0^1 \frac{dx}{1+x} = [\ln(1+x)]_0^1 = \ln 2 - \ln 1$

$= \ln 2 - 0 = \ln 2$

40 $\because \int_{\pi/2}^{3\pi/2}[2\sin x]dx = \int_{\pi/2}^{5\pi/6}[2\sin x]dx + \int_{5\pi/6}^{\pi}[2\sin x]dx$

$+\int_{\pi}^{7\pi/6}[2\sin x]dx + \int_{7\pi/6}^{3\pi/2}[2\sin x]dx$

$= \int_{\pi/2}^{5\pi/6} 1.dx + 0 - \int_{\pi}^{7\pi/6} 1.dx - 2\int_{7\pi/6}^{3\pi/2} 1.dx$

$= \frac{\pi}{3} - \frac{\pi}{6} - \frac{2\pi}{3} \quad = \frac{\pi}{2}$

$\left(\because 2\sin x \text{ is decreasing function in} \left(\frac{\pi}{2}, \frac{3\pi}{2}\right)\right)$

41. Use formula $\int_0^n [x]dx = \frac{n(n-1)}{2}$

42. Use formula $\int_0^{\frac{\pi}{2}} \frac{f(\sin x)}{f(\sin x) + f(\cos x)} dx = \frac{\pi}{4}$

43. Let

$I = \int_0^{1/4} \frac{dx}{1+(\cot 2\pi x)^{\sqrt{2}}} = \frac{1}{2\pi}\int_0^{\pi/2} \frac{dx}{1+(\cot x)^{\sqrt{2}}}$

$= \frac{1}{2\pi}\int_0^{\pi/2} \frac{(\sin x)^{\sqrt{2}}}{(\sin x)^{\sqrt{2}} + (\cos x)^{\sqrt{2}}} dx$

$= \frac{1}{2\pi}\left\{\frac{\frac{\pi}{2} - 0}{2}\right\} = \frac{1}{8}$

and let $I_1 = \int_a^b \frac{f(\sin x)}{f(\sin x) + f(\cos x)} dx$ ..... (1)

$I_1 = \int_a^b \frac{f(\sin(a+b-x))}{f(\sin(a+b-x)) + f(\cos(a+b-x))} dx$

$= \int_a^b \frac{f(\cos x)}{f(\cos x) + f(\sin x)} dx$ ...... (2)

Adding equations (1) and (2), then

$2I_1 = \int_0^b dx = b - a \qquad \therefore I_1 \frac{(b-a)}{2}$

44. |sinx| is periodic function with period π

45. $\because I = \int_0^{2\pi} \sin^3 x\, dx = \int_0^{2\pi} (1-\cos^2 x)\sin x\, dx$

Put $\cos x = t \Rightarrow \sin x\, dx = -dt$

Then, $I = \int_1^1 (1-t^2)(-dt) = 0$

46. The function $\frac{|x|}{x}$ is undefined at $x = 0$

$\therefore \int_{-1}^1 \frac{|x|}{x} dx \qquad = \int_{-1}^0 \frac{|x|}{x} dx + \int_0^1 \frac{|x|}{x} dx$

$= \int_{-1}^0 \frac{-x}{x} dx + \int_0^1 \frac{x}{x} dx \qquad = -\int_{-1}^0 dx + \int_0^1 dx$

$= -(0-(-1)) + (1-0) \quad = 0$

47. $\because \cos x$ is continuous in $[0,t]$

$\therefore \int_0^t \cos x\, dx = [\sin x]_0^t = \sin t - \sin 0$

48. $\therefore \int_1^6 \{x+5\}^2 dx = \int_0^5 \{x+6\}^2 dx$ (by property)

$= \int_0^5 \{x\}^2 dx = 5\int_0^1 \{x\}^2 dx$

$(\because [.] \text{ is periodic with period } 1) = 5\int_0^1 x^2 dx \quad = \frac{5}{3}$

49. $\therefore f(x)$ is continuous in $[0,2]$

$\therefore \int_0^2 f(x)dx = \int_0^1 f(x)dx + \int_1^2 f(x)dx$

$\therefore \int_0^1 x^2 dx + \int_1^2 \sqrt{x} dx \qquad = \frac{1}{3} + \frac{2}{3}(2^{3/2} - 1)$

$= \frac{1}{3} + \frac{4\sqrt{2}}{3} - \frac{2}{3} \qquad = \left(\frac{4\sqrt{2}-1}{3}\right)$

50. $\because 0 < x < 1$

then $x > x^2 \Rightarrow -x < -x^2$

$\Rightarrow e^{-x} < e^{-x^2}$

$\Rightarrow e^{-x}\cos^2 x < e^{-x^2}\cos^2 x$

Then, $\int_0^1 e^{-x}\cos^2 x\,dx < \int_0^1 e^{-x^2}\cos^2 x\,dx$ and if $f(x) \ge g(x)$

Then, $\int_a^b f(x)\,dx \ge \int_a^b g(x)\,dx$

51. For $x > 0$; $\sin x < x$ $\Rightarrow \frac{\sin x}{x} < 1$

$\Rightarrow \int_0^{\pi/2} \frac{\sin x}{x}dx < \int_0^{\pi/2} 1.dx$

$\Rightarrow \int_0^{\pi/2} \frac{\sin x}{x}dx < \frac{\pi}{2}$

and $\lim_{x\to 0}\frac{\sin x}{x} = 1$

52. $\because f(x) = x - x^2 + 1$

$\therefore f'(x) = 1 - 2x$

$f'(x) > 0 \Rightarrow 1 - 2x > 0 \Rightarrow x < \frac{1}{2}$ and

$f'(x) < 0 \Rightarrow 1 - 2x < 0 \Rightarrow x > \frac{1}{2}$

$\Rightarrow f(x)$ is increasing in (0, 1/2) and decreasing in (1/2, 1)

Now, $g(x) = \max\{f(t); 0 \le t \le x\}$

$$= \begin{cases} x - x^2 + 1, & 0 \le x \le \frac{1}{2} \\ \frac{1}{2} - \frac{1}{4} + 1, & \frac{1}{2} \le x \le 1 \end{cases}$$

$$= \begin{cases} x - x^2 + 1, & 0 \le x \le \frac{1}{2} \\ \frac{5}{4}, & \frac{1}{2} \le x \le 1 \end{cases}$$

$\therefore \int_0^1 g(x)\,dx = \int_0^{1/2}\left(x - x^2 + 1\right)dx + \int_{1/2}^1 \frac{5}{4}dx$

$= \left[\frac{x^2}{2} - \frac{x^3}{3} + x\right]_0^{1/2} + \frac{5}{4}[x]_{1/2}^1$

$= \left(\frac{1}{8} - \frac{1}{24} + \frac{1}{2}\right) + \frac{5}{4}\left(1 - \frac{1}{2}\right)$

$= \frac{1}{8} - \frac{1}{24} + \frac{1}{2} + \frac{5}{8}$

$= \frac{3 - 1 + 12 + 15}{24} = \frac{29}{24}$

53. By using formula

54. Standard result

55. A---- $\frac{1}{4 + x^2}$ is even function

B---- Given $= \left(\sec^{-1} x\right)_1^2$

C--- Use formula $\int_0^{2a} f(x)dx = 0$ if f(2a-x) = -f(x)

58. Use formula $\int_0^{\frac{\pi}{2}} \sin^n x\,dx = \frac{n-1}{n}.\frac{n-3}{n-2}.....\frac{1}{2}.\frac{\pi}{2}$ if n is even

59. $A \to \int_0^3 [x]dx = \frac{3(3-1)}{2}$ $B \to x^3$ is odd function

$C \to \int_0^3 \left(x - [x]\right)dx = 3\int_0^1 xdx$

$D \to \int_{-1}^1 |x|\,dx = 2\int_0^1 xdx$